

Detecting Cytokine Cascade Direction from Latent Geometry and Training Dynamics

Methods, Validation, and Comparison

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1 Biological Motivation

Cytokine cascades are directional: cytokine A activates cells of type X , which produce cytokine B as a downstream effector. The goal is to infer these directed relationships — and identify the relay cell type X — from single-cell RNA-seq data, without prior knowledge of which cytokines cascade into which.

The key observation is the following. In the Oesinghaus et al. dataset, cells are cultured in pseudo-tubes: closed bags of ≈ 480 PBMCs treated with a single cytokine for 24 hours. If cascade $A \rightarrow B$ exists, cells inside an A -treated tube will have begun responding to *endogenously produced* B — because cells that were directly activated by A start secreting B into the same tube. As a result, an A -tube contains a mixed signal: a strong direct response to A and a weaker secondary response to B .

Crucially, the reverse is not true: a B -treated tube contains no endogenously produced A . This **directional asymmetry** — A -tubes look partially like B -tubes, but not vice versa — is the cascade signal we aim to detect.

2 System Overview

The full pipeline is illustrated in Figure 1. A residual MLP encoder maps individual cells to 128-dimensional embeddings; an attention module aggregates these into a bag-level representation; a linear classifier predicts the cytokine label. The cascade detection methods operate on the encoder embeddings after applying the PBS-RC transform described in Section 4.

3 Mathematical Setting

Cells and expression. Let $\mathcal{I}$ be the set of all cells. Each cell $i \in \mathcal{I}$ has a gene expression vector $x_i \in \mathbb{R}^G$ (log-normalised, $G = 4,000$ HVGs), a cytokine label $c(i) \in \mathcal{C}$ (where C_0 denotes PBS), a cell-type label $\tau(i) \in \mathcal{T}$ ($M = 18$ annotated PBMC types), and a donor label $d(i) \in \mathcal{D}$ ($|\mathcal{D}| = 12$).

Encoder. A residual MLP $f_\theta : \mathbb{R}^G \rightarrow \mathbb{R}^d$ ($d = 128$) maps each cell to a dense embedding $h_i = f_\theta(x_i)$. f_θ is pre-trained by cell-type classification (Stage 1); it has never seen cytokine labels, so all cytokine signal in h_i is emergent.

Donors. The dataset is split at the donor level: $\mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}}$, with $\mathcal{D}_{\text{train}} = \{\text{D1, D4–D12}\}$ and $\mathcal{D}_{\text{val}} = \{\text{D2, D3}\}$. All centroids are computed on $\mathcal{D}_{\text{train}}$ only.

Cell sets. For cytokine $A \in \mathcal{C}$ and cell type $T \in \mathcal{T}$: $\mathcal{S}_{A,T} = \{i \in \mathcal{I} : c(i) = A, \tau(i) = T, d(i) \in \mathcal{D}_{\text{train}}\}$.

AB-MIL Cytokine Cascade Inference Pipeline

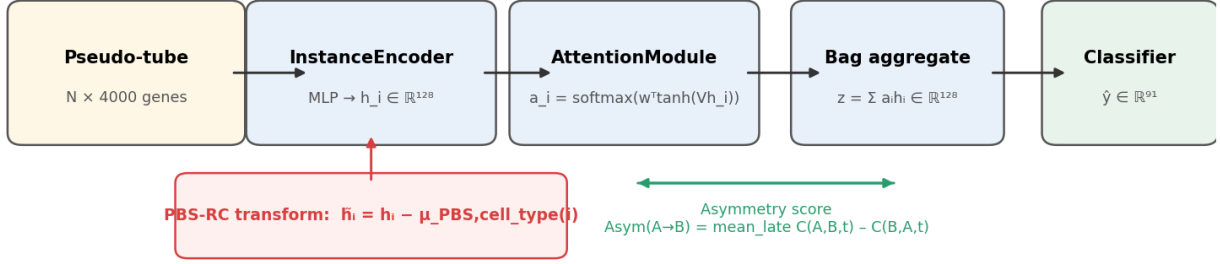


Figure 1: **AB-MIL Cytokine Cascade Inference Pipeline.** Each pseudo-tube ($N \times 4000$ genes) is passed through an InstanceEncoder (MLP, $h_i \in \mathbb{R}^{128}$), attention aggregation, and a 91-class classifier. The PBS-RC transform (red box) subtracts the cell-type-specific resting-state centroid from each embedding before computing cascade scores. The green arrow shows where asymmetry scores are read off from the attention-weighted centroid trajectories.

4 The PBS-RC Transform

4.1 The Composition Problem

The dominant axis of variation in $\mathbb{R}^d$ is cell type: every NK cell lands near the NK cluster regardless of cytokine treatment. Figure 2 illustrates this: in raw embedding space, cytokine centroids nearly coincide and cell-type identity overwhelms any cytokine-induced displacement. After the PBS-RC transform, all cell types are centred at their own resting state and within-cluster displacements become visible.

A naïve comparison of cytokine centroids $\bar{h}_A$ conflates cell-state changes with cell-type composition differences between conditions. Even the within-tube displacement ($\mu_{A,T} - \mu_A$) fails without PBS subtraction: if cytokine A causes NK proliferation, μ_A shifts toward the NK cluster even if no NK cell changed its transcriptional state, producing a spurious bias. The PBS-RC transform resolves this.

4.2 PBS-RC Embedding

Definition 1 (PBS reference centroid). *For each cell type $T \in \mathcal{T}$, the PBS reference centroid is*

$$\mu_{\text{PBS},T} = \frac{1}{|\mathcal{S}_{C_0,T}|} \sum_{i \in \mathcal{S}_{C_0,T}} h_i \in \mathbb{R}^d.$$

Definition 2 (PBS-RC embedding). *The PBS-Relative Centroid embedding of cell i is $\tilde{h}_i = h_i - \mu_{\text{PBS},\tau(i)} \in \mathbb{R}^d$.*

By construction, $\tilde{h}_i = \mathbf{0}$ whenever cell i occupies the same position as the average resting-state cell of type $\tau(i)$. Because the subtraction is cell-type-specific, it genuinely reshapes the geometry rather than shifting all cells by the same global offset.

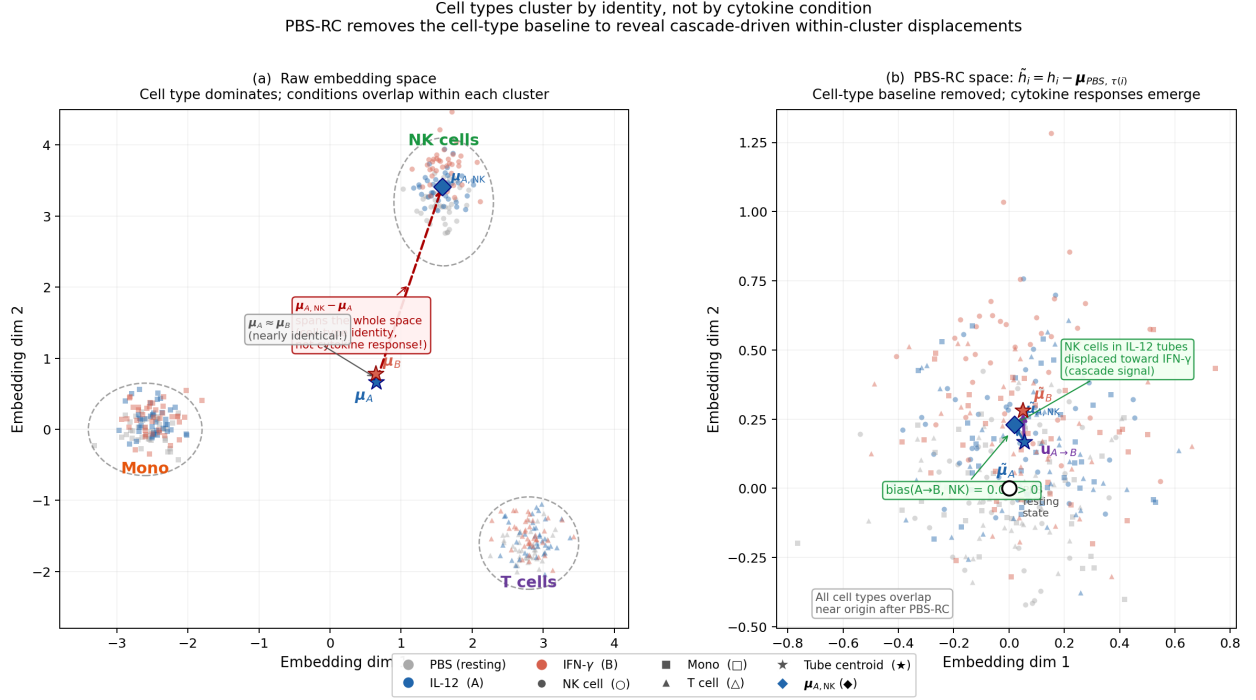


Figure 2: **PBS-RC resolves the composition confound.** (a) Raw embedding space: cell-type identity dominates; NK, Mono, and T cells from IL-12, IFN- γ , and PBS tubes all overlap within their respective clusters. IL-12 and IFN- γ tube centroids are indistinguishable at this scale. (b) PBS-RC space $\tilde{h}_i = h_i - \mu_{\text{PBS}, \tau(i)}$: all cell types are centred at the origin; NK cells in IL-12 tubes are visibly displaced toward IFN- γ space (cascade signal). The bias measurement along $\hat{u}_{A \rightarrow B}$ is shown.

4.3 Cytokine Centroids in PBS-RC Space

For cytokine $A \in \mathcal{C} \setminus \{C_0\}$:

$$\mu_A = \frac{1}{|\mathcal{S}_A|} \sum_{i \in \mathcal{S}_A} \tilde{h}_i, \quad \mu_{A,T} = \frac{1}{|\mathcal{S}_{A,T}|} \sum_{i \in \mathcal{S}_{A,T}} \tilde{h}_i.$$

The deviation $(\mu_{A,T} - \mu_A)$ measures how much cell type T is displaced within A -tubes relative to the average A -tube cell. Figure 3 summarises the geometric interpretation.

Figure 4 shows the same transform applied to real encoder embeddings.

5 Static Cascade Direction Methods

5.1 Directional Bias

For cytokines $A, B \in \mathcal{C} \setminus \{C_0\}$ with $A \neq B$, let

$$\mathbf{u}_{A \rightarrow B} = \frac{\mu_B - \mu_A}{\|\mu_B - \mu_A\|_2}$$

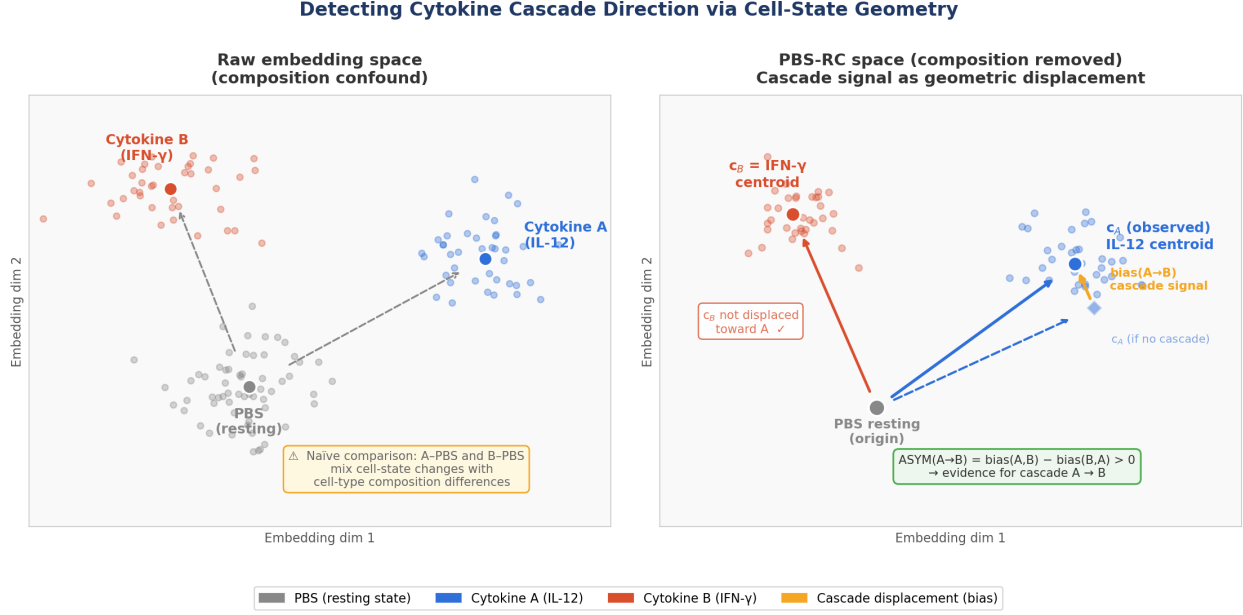


Figure 3: **Cascade detection via cell-state geometry.** *Left:* Raw embedding space. Cell-type identity dominates; cytokine tube centroids nearly coincide. *Right:* PBS-RC space. All cell types are centred at the origin; the observed IL-12 centroid c_A is displaced toward the IFN- γ centroid c_B , whereas the non-cascading direction shows no displacement. The asymmetry score $\text{ASYM}(A \rightarrow B) = \text{bias}(A, B) - \text{bias}(B, A) > 0$ is evidence for cascade $A \rightarrow B$.

be the unit vector pointing from A 's centroid to B 's centroid. The **directional bias** of cell type T in A -tubes toward B is:

$$\text{bias}(A \rightarrow B, T) = (\mu_{A,T} - \mu_A) \cdot \mathbf{u}_{A \rightarrow B} \in \mathbb{R}.$$

Figure 5 shows a worked 2D example: $\text{bias}(\text{IL-12} \rightarrow \text{IFN-}\gamma) = 0.80$ and $\text{bias}(\text{IFN-}\gamma \rightarrow \text{IL-12}) = 0.625$, giving $\text{ASYM} = 0.175 > 0$.

5.2 Why This Captures Cascade Direction

Suppose cascade $A \rightarrow B$ exists via relay cell type X :

1. Cytokine A is added to the pseudo-tube.
2. Cells of type X are primary responders: directly activated by A , they begin secreting B autocrinally. Their state reflects *both* A and B .
3. All other cells receive B paracrinally and shift toward B -space.

X receives a double signal and shifts *more* than the average tube cell: $\text{bias}(A \rightarrow B, X) > 0$. In a B -treated tube there is no endogenous A ; X responds to B like every other cell type, so $\text{bias}(B \rightarrow A, X) \approx 0$.

5.3 Legacy Asymmetry Score and Its Structural Problems

The legacy aggregate score was:

$$\text{ASYM}(A \rightarrow B) = \max_{T \in \mathcal{T}} [\text{bias}(A \rightarrow B, T) - \text{bias}(B \rightarrow A, T)].$$

Cascade signal: NK cells in IL-12 tubes shift toward IFN- γ space (bias > 0); Mono cells do not

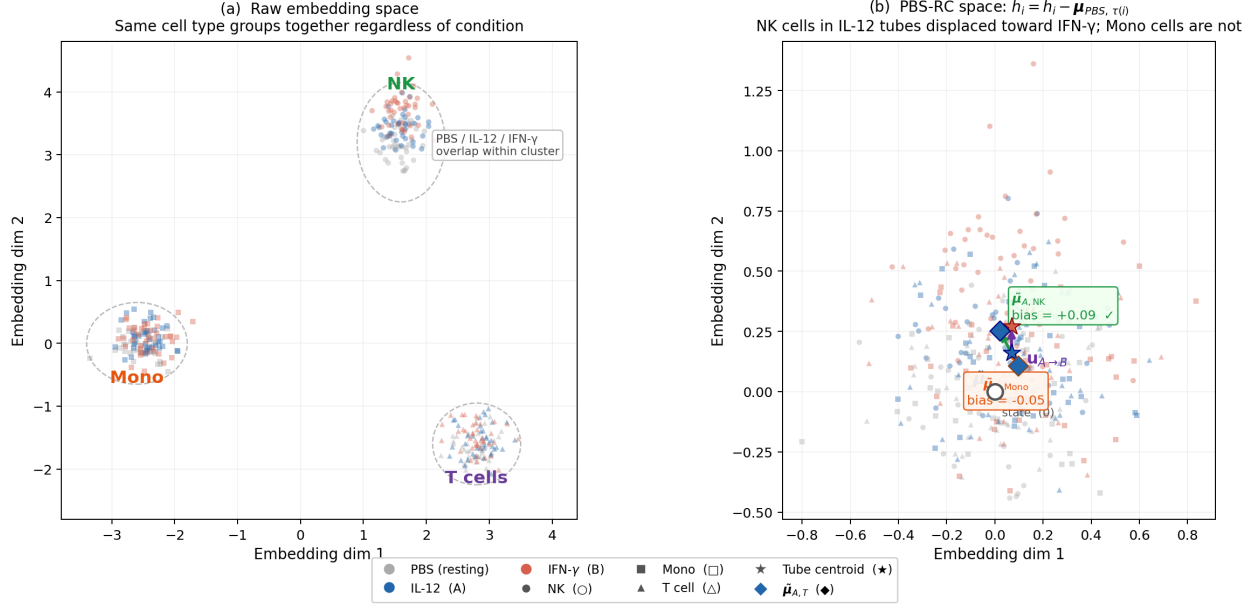


Figure 4: **PBS-RC reveals within-cluster cytokine displacements.** *Left:* Raw encoder space. Cell-type identity dominates; IL-12 and IFN- γ tube centroids nearly coincide, and the NK displacement $\mu_{A,NK} - \mu_A$ spans the whole space. *Right:* After the PBS-RC transform, cytokine centroids separate, and NK cells in IL-12 tubes are visibly displaced toward IFN- γ space (bias = +0.09), while Monocytes show no displacement (bias = -0.05).

The max over cell types is motivated by the fact that a cascade acts through a specific relay; the argmax directly identifies X^* .

Despite being well-motivated, the subtraction introduces two structural defects.

Contamination. Since $\hat{u}_{B \rightarrow A} = -\hat{u}_{A \rightarrow B}$, expanding the subtraction gives:

$$\text{bias}(A \rightarrow B, T) - \text{bias}(B \rightarrow A, T) = \underbrace{(\mu_{A,T} - \mu_A) \cdot \hat{u}_{A \rightarrow B}}_{\text{forward (wanted)}} + \underbrace{(\mu_{B,T} - \mu_B) \cdot \hat{u}_{A \rightarrow B}}_{\text{contaminant}}.$$

A cell type T that responds strongly *directly* to B inflates $\text{ASYM}(A \rightarrow B)$ even when no cascade through T exists.

Forced antisymmetry. $\text{ASYM}(A \rightarrow B) = -\text{ASYM}(B \rightarrow A)$ identically. The method cannot distinguish a genuine directed cascade from an algebraic sign flip.

Figure 6 illustrates the method and its failure modes.

Figure 7 shows the full 91×91 legacy asymmetry matrix.

5.4 Refined Static Pipeline: Per-Donor Wilcoxon

The refined pipeline keeps the same geometric intuition but replaces the contaminated subtraction with two fully independent one-sided statistical tests.

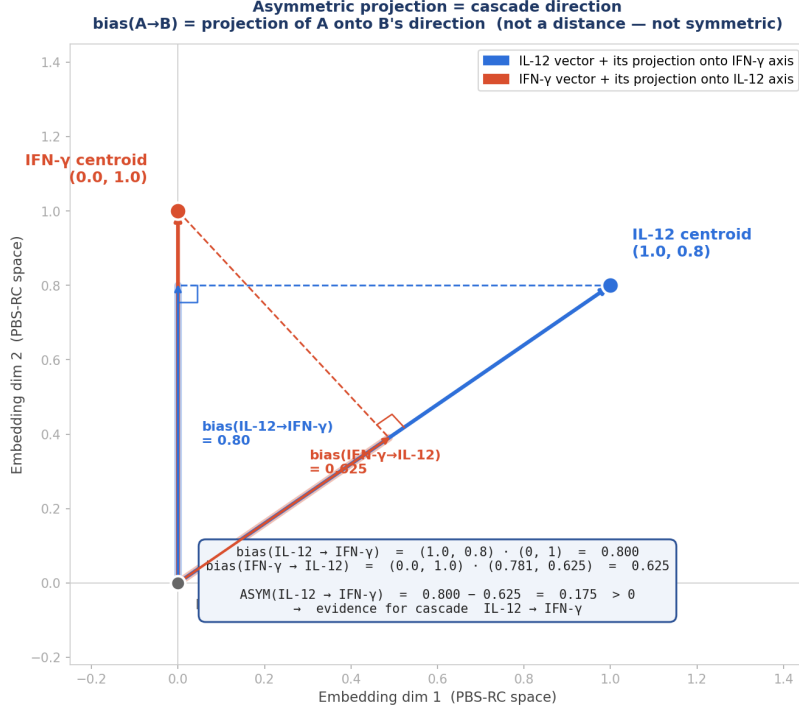


Figure 5: **Directional bias is an asymmetric projection.** IL-12 centroid (1.0, 0.8) and IFN- γ centroid (0.0, 1.0) in a 2D PBS-RC space. The blue vector is IL-12’s deviation projected onto $\hat{u}_{A \rightarrow B}$ (bias = 0.80); the red vector is IFN- γ ’s projection onto the same axis (bias = 0.625). The asymmetry 0.175 > 0 indicates IL-12 $\rightarrow$ IFN- γ . The key property: bias is a projection, not a distance — it is not symmetric.

Step 1 — Per-donor PBS-RC. For every cell type T and training donor d : $\tilde{h}_i = h_i - \mu_{\text{PBS}, \tau(i)}^{(d)}$, using donor-specific PBS centroids.

Step 2 — Per-donor forward projection.

$$b_{\text{fwd}}^{(d)}(A \rightarrow B, T) = (\tilde{\mu}_{A, T}^{(d)} - \tilde{\mu}_A) \cdot \hat{u}_{A \rightarrow B},$$

where $\tilde{\mu}_A = \frac{1}{|\mathcal{D}_{\text{train}}|} \sum_d \tilde{\mu}_A^{(d)}$. No reverse term is subtracted anywhere. The “reverse” of (A, B) is the independent forward test for (B, A) .

Step 3 — Statistical testing. One-sided Wilcoxon signed-rank test across $d \in \mathcal{D}_{\text{train}}$ (null: median ≤ 0) for each (A, B, T) . Two-stage correction: Bonferroni across cell types within each ordered pair (relay search), then Benjamini–Hochberg FDR ($\alpha = 0.05$) across all $K(K-1) = 8,010$ ordered pairs.

Cascade call. $\text{fwd_sig} = \exists T : p_{\text{Bonf}}(A \rightarrow B, T) \leq \alpha$, $\text{rev_sig} = \exists T : p_{\text{Bonf}}(B \rightarrow A, T) \leq \alpha$, producing one of: A $\rightarrow$ B, B $\rightarrow$ A, shared, or none.

6 Centroid Trajectory Method

The static methods use a single frozen-encoder snapshot. An alternative approach exploits the *dynamics* of fine-tuning: if cascade $A \rightarrow B$ exists, the cell-type centroids in A -tubes should systematically drift toward B 's centroid as the model is fine-tuned to discriminate cytokine labels.

6.1 Training Protocol

Figure 8 shows the three-stage training protocol and the trajectory concept.

At each logged epoch t , the **attention-weighted centroid** for cytokine C , cell type T , donor d is:

$$\tilde{\mu}_{C,T}^{(d)}(t) = \frac{\sum_{i \in \mathcal{S}_{C,T}^{(d)}} a_i(t) \tilde{h}_i(t)}{\sum_{i \in \mathcal{S}_{C,T}^{(d)}} a_i(t)},$$

where $a_i(t)$ are the MIL attention weights and $\tilde{h}_i(t)$ is the PBS-RC embedding at epoch t . The trajectory score for pair (A, B) , cell type T , donor d is the OLS slope of $\{b_{\text{fwd}}^{(d)}(A \rightarrow B, T, t)\}_{t=10,20,\dots,100}$. A positive slope means relay cell type T drifts progressively toward B 's centroid.

6.2 Generalization Check

Figure 9 confirms the model learns cytokine-specific programs that generalise to held-out donors.

6.3 NK Slope Analysis for the Positive Control (IL-12 $\rightarrow$ IFN- γ)

The IL-12 $\rightarrow$ IFN- γ cascade (mediated by NK cells) is the best-documented PBMC cascade in the dataset. Figure 10 shows the per-donor NK trajectory slopes across 10 seeds.

Figure 11 aggregates the per-donor slopes to the seed level.

6.4 PCA Trajectory Visualization

Figures 12 and 13 show the centroid trajectories in the cascade-relevant 2D PCA space.

6.5 Benchmark Recovery

The trajectory method was evaluated against 11 literature-curated cascade pairs (9 positive + 2 controls). Figure 14 shows the result.

The trajectory method achieves **11/11 pairs** $\times$ **10/10 seeds** (100% recall) on the benchmark set. This is the strongest validation result to date, substantially improving on the static methods (see Section 8.4).

7 Experimental Setup

7.1 Batch Construction

The full dataset contains 12 donors. For each cytokine c_i ($i = 0, \dots, 90$), one tube is selected from donor $(i + \delta) \bmod 12$. Two mutually exclusive batches:

- **Batch 0** ($\delta = 0$, seeds 1–10, 42, 123).

- **Batch 1** ($\delta = 1$, seeds 11–22).

Every cytokine is represented by a *different* donor across the two batches, so the training cells are mutually exclusive.

7.2 Pre-Registered Controls

Before inspecting any results, four cytokine pairs were designated as controls (Table 1). Figure 15 shows the PBS-RC rank percentile for these pairs before and after the transform.

Table 1: Pre-registered pairs of interest.

Direction	Pair	Role	Expected signal
$A \rightarrow B$	IL-12 $\rightarrow$ IFN- γ	Positive control	High score; $A \rightarrow B > B \rightarrow A$
$B \rightarrow A$	IFN- $\gamma \rightarrow$ IL-12	Positive control (reverse)	Lower score than forward
$A \rightarrow B$	IL-6 $\rightarrow$ IL-10	Negative control	Mid-range, symmetric with reverse
$B \rightarrow A$	IL-10 $\rightarrow$ IL-6	Negative control (reverse)	Similar rank to forward

8 Results: Static Methods

8.1 Within-Batch Seed Stability

A seed is *stable* if its Spearman ρ with the batch ensemble mean is ≥ 0.70 . Figures 16a and 16b show per-seed ρ for Batch 0 and Batch 1.

8.2 Cross-Batch Reproducibility

Figure 17 shows the cross-batch scatter for the legacy method. The two batches use entirely disjoint training cells.

Top discovered pairs. Two cytokines dominate the top of both batch rankings: **IL-27** (documented downstream targets include IL-22 and TNF- α) and **IFN- β** (type-I interferon with downstream ISG programme activation). Figure 18 shows the top pairs from Batch 0 and per-seed stability.

8.3 Legacy vs. Refined: Stability and Cross-Batch Comparison

Both methods were evaluated on the full 24-run corpus (Batch 0 and Batch 1, 12 seeds each). Figure 19 compares per-seed ρ side by side.

Figure 20 shows the cross-batch scatter for both methods under different pair-set filters.

Table 2 reports cross-batch ρ and top-50 overlap under four pair-set filters. Row 3 is the critical diagnostic: restricting to the top 50% by legacy magnitude, legacy ρ *drops* to 0.620 while refined ρ on the same pairs *rises* to 0.867. The legacy method’s highest-magnitude pairs are its *least* reproducible.

8.4 Benchmark Comparison: Static Methods

Figure 21 evaluates both static methods against the 11 literature-curated benchmark pairs.

Table 2: Cross-batch Spearman ρ and top-50 overlap under four pair-set filters. “Top-50 overlap” counts how many of the top-50 pairs by Batch-0 score also appear in the top-50 of Batch-1. Row 3 is the critical diagnostic.

Filter	Legacy ρ	Refined ρ	Legacy overlap	Refined overlap
All pairs ($n = 8,010$)	0.772	0.815	25/50	3/50
Top 50% by refined score ($n = 4,005$)	0.762	0.835	28/50	10/50
Top 50% by legacy score ($n = 4,005$)	0.620	0.867	25/50	9/50
Active pairs (refined > 0 , $n = 7,100$)	0.772	0.813	27/50	3/50

9 Limitations and Caveats

1. **Encoder is cell-type supervised, not cytokine supervised.** Cytokine signals are encoded incidentally as deviations within cell-type clusters. If cytokine A induces a programme identical to resting-state variation in cell type T , it will not appear as a displacement in PBS-RC space.
2. **Small effective donor sample.** All centroids are computed from $|\mathcal{D}_{\text{train}}| = 10$ donors. Pseudo-tubes from the same donor are correlated; effective $n = 10$, not the number of cells. Statistical power is limited and the donor-level Wilcoxon test is the correct unit of analysis.
3. **24-hour snapshot.** All tubes are measured at a single 24-hour time point. Cytokines with slow induction kinetics may not have produced sufficient secondary B for the effect to appear. A positive score is consistent with a cascade but could also reflect simultaneous shared programme activation.
4. **Bimodal seed split in the trajectory method.** 4/10 seeds show negative NK slopes for the IL-12 $\rightarrow$ IFN- γ positive control (Figure 11), suggesting multiple attractor basins in the fine-tuning landscape. The source of this split is not yet understood.
5. **Top-50 overlap remains low for static methods.** Even under the refined method, top-50 cross-batch overlap reaches only 10/50. Monotone improvement with filtering suggests a threshold problem rather than absent signal, but stricter FDR thresholds or more seeds are needed to verify.

10 Conclusions

1. **The PBS-RC transform is necessary.** Without it, cell-type composition confounds cytokine centroid comparisons. After the transform, within-cluster cytokine-induced displacements become visible and interpretable.
2. **Static methods are reproducible above threshold.** Within-batch stability is high (10–11/12 seeds) and cross-batch ρ exceeds the pre-specified 0.70 threshold on disjoint training data (legacy 0.766, refined 0.815).
3. **Legacy contamination is exposed.** Restricting to the top 50% by legacy magnitude drops legacy ρ to 0.620 while raising refined ρ to 0.867 on the same pairs, confirming that contamination drives the largest legacy scores.
4. **The centroid trajectory method achieves 100% benchmark recall.** All 11 literature-curated cascade pairs are correctly recalled across all 10 seeds, compared to partial recall by both static methods. This is the strongest validation result to date.
5. **Pre-registered controls behave as expected.** IL-12 $\rightarrow$ IFN- γ ranks consistently in the upper decile with correct directionality; IL-6/IL-10 is correctly reclassified as a shared-pathway pair after PBS-RC.

6. **IL-27 and IFN- β emerge as robust cascade sources.** Their top-pair lists overlap substantially between independent batches and match documented downstream biology.

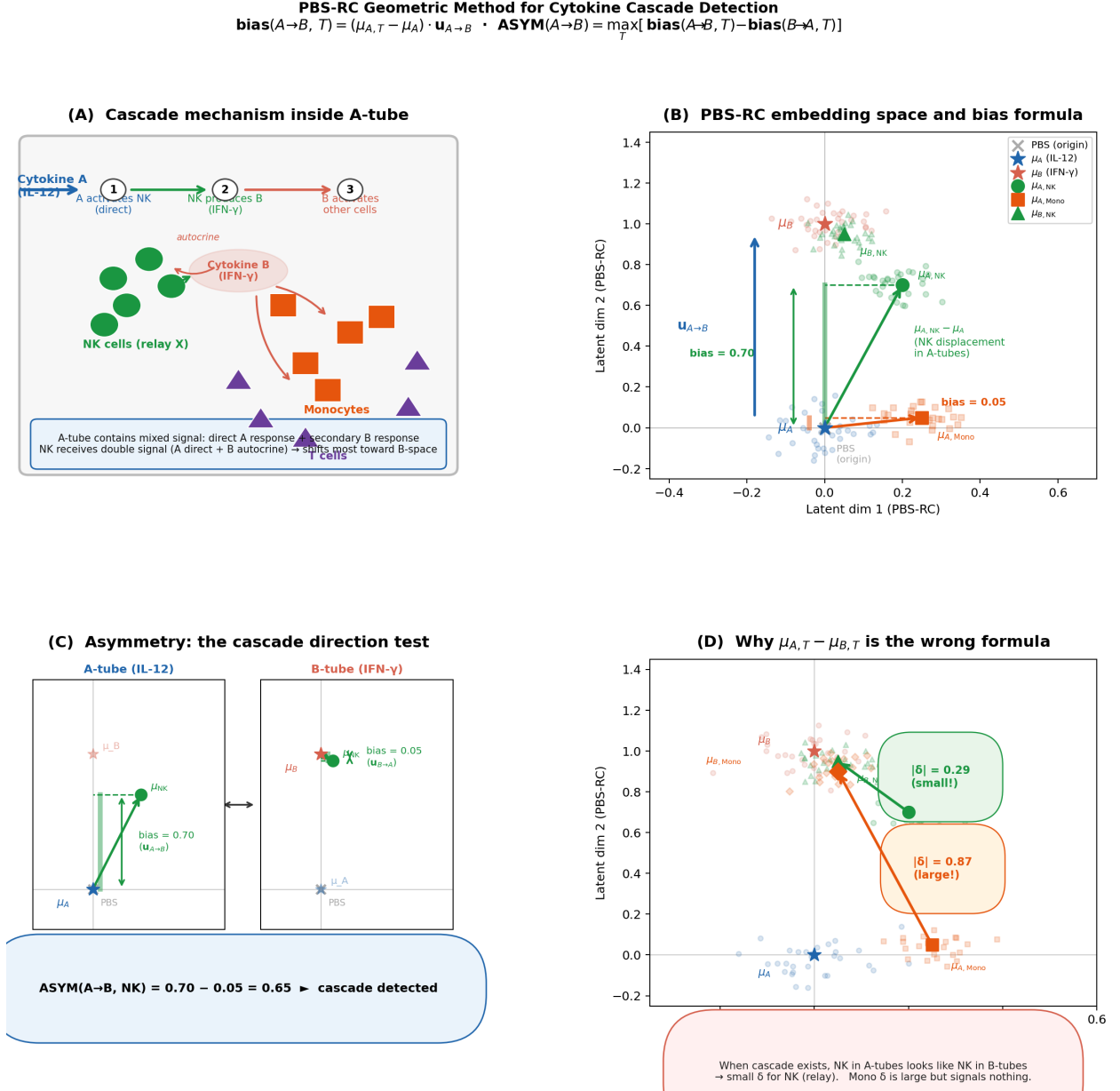


Figure 6: **PBS-RC geometric method for cascade detection (four-panel summary).** (A) Biological mechanism: cytokine A directly activates relay cell type X (NK cells), which secrete B autocrinally. (B) PBS-RC space: NK cells in A -tubes are strongly displaced toward B (bias = 0.70); Monocytes barely deviate (bias = 0.05). (C) Asymmetry test: large forward bias minus small reverse bias gives $\text{ASYM}(A \rightarrow B) = 0.65$, identifying NK as the relay X^* . (D) Why $\mu_{A,T} - \mu_{B,T}$ is the wrong formula: when a cascade exists, NK in A -tubes resembles NK in B -tubes (small $|\delta|$, invisible), while Monocytes differ for unrelated reasons (large $|\delta|$, spurious). The naïve formula inverts the ranking.

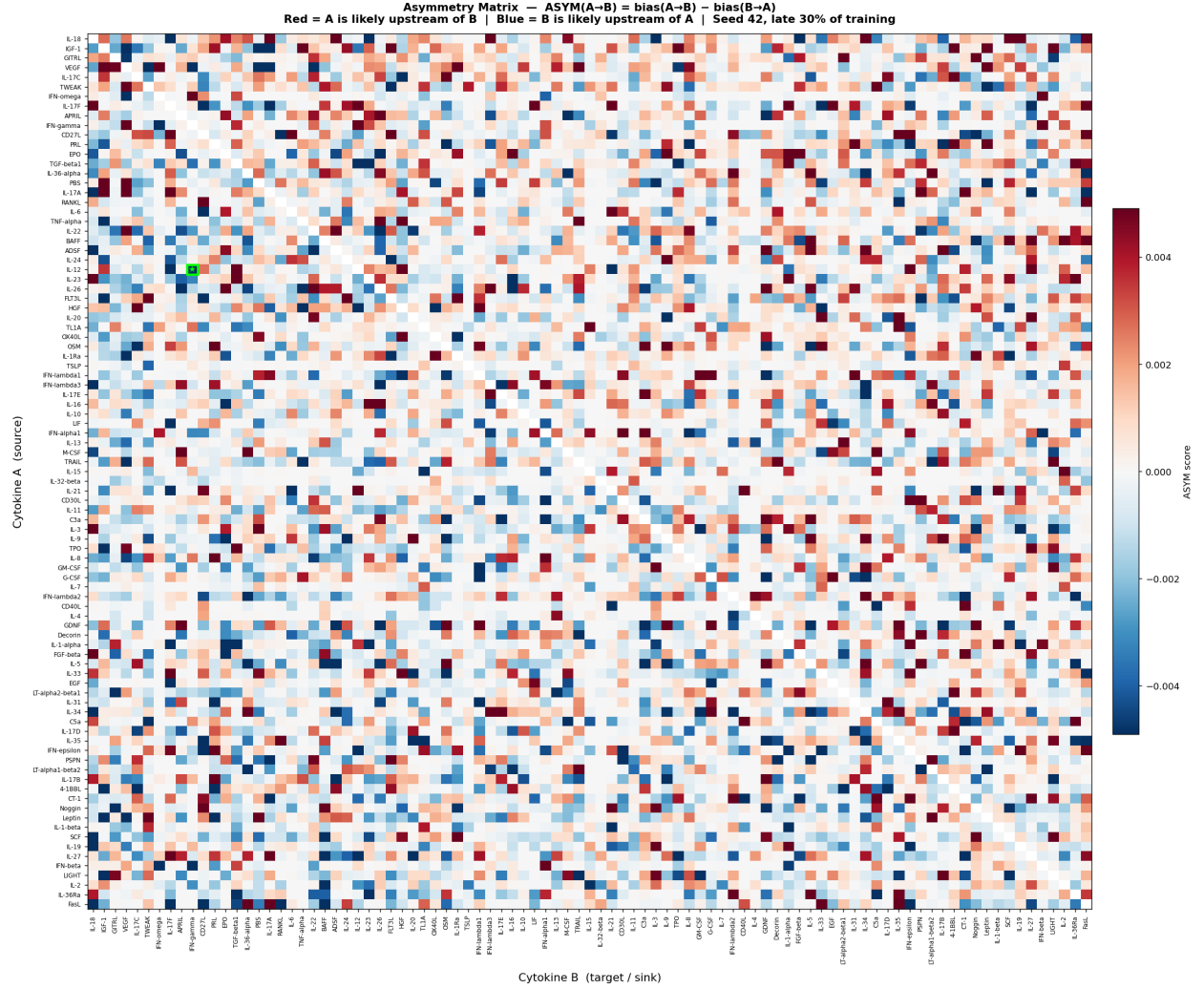


Figure 7: **Full legacy asymmetry matrix (91×91 ordered pairs).** $ASYM(A \rightarrow B) = \text{bias}(A, B) - \text{bias}(B, A)$. Red = A likely upstream of B; Blue = B likely upstream. Seed 42, late 30% of training. Several cytokines (e.g. IL-27, IFN- β) show consistent asymmetry rows, suggesting they are common cascade sources. The anti-symmetric structure (each row is the sign-flip of the corresponding column) is a structural artefact of the legacy formula.

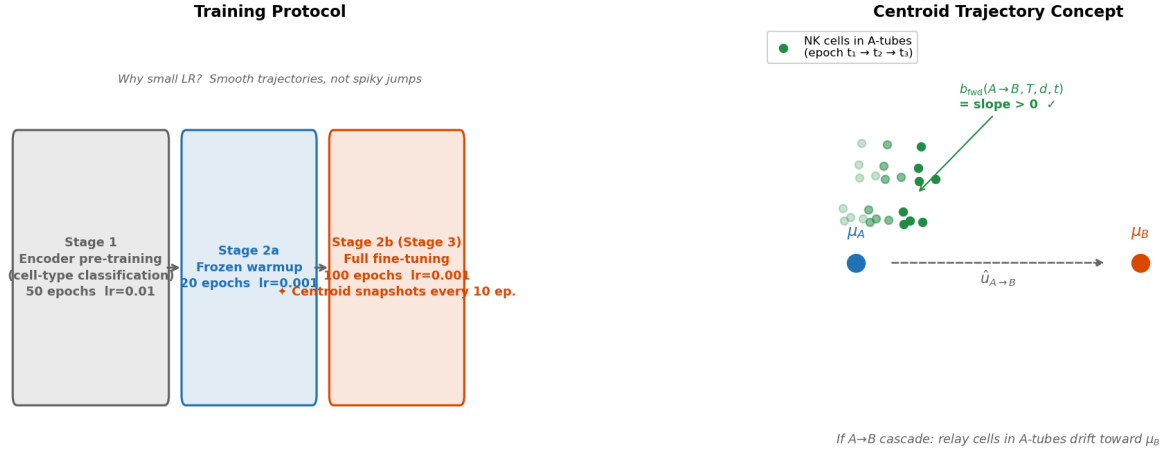


Figure 8: **Training protocol and centroid trajectory concept.** *Left:* Three stages. Stage 1: encoder pre-training on cell-type labels (50 epochs, $lr = 0.01$). Stage 2a: frozen-encoder MIL warmup (20 epochs, $lr = 0.001$). Stage 2b/3: full fine-tuning of both encoder and MIL weights (100 epochs, $lr = 0.001$) with attention-weighted centroid snapshots logged every 10 epochs. Small LR chosen for smooth trajectories, not spiky jumps. *Right:* Trajectory concept. If $A \rightarrow B$ cascade exists, attention-weighted centroids of relay cell type T in A -tubes (green dots, $t_1 \rightarrow t_2 \rightarrow t_3$) drift toward μ_B along the $\hat{u}_{A \rightarrow B}$ axis, giving a positive b_{fwd} slope.

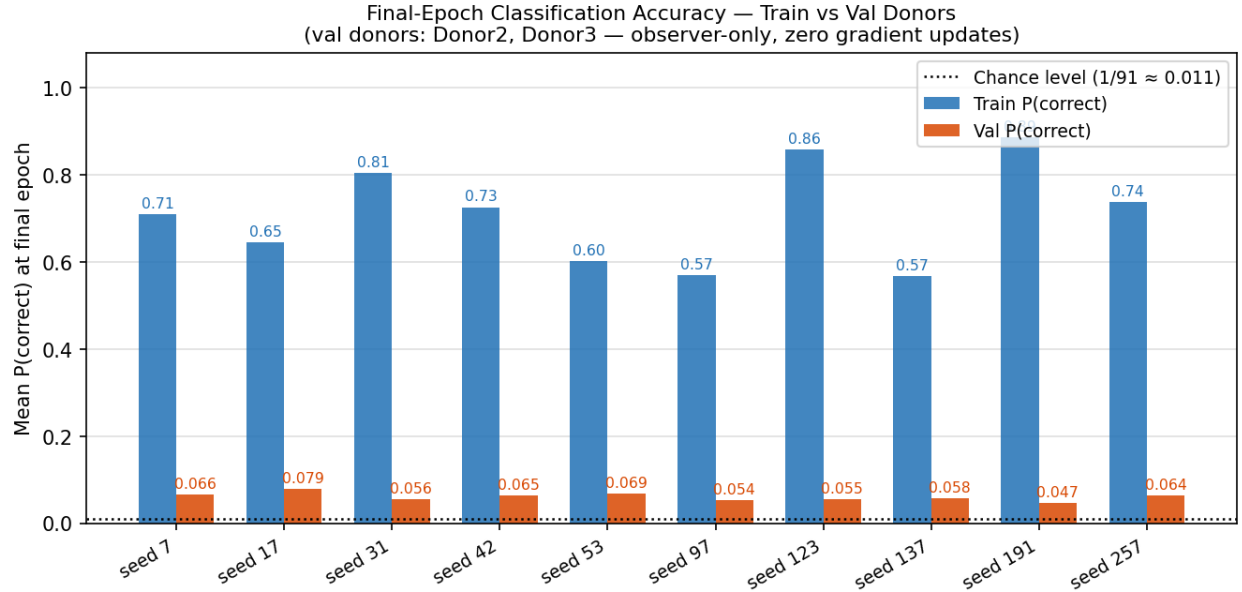


Figure 9: **Final-epoch classification accuracy across 10 seeds.** Train donors (D1, D4–D12): mean $P(\text{correct})$ ranges from 0.57 to 0.86. Val donors (D2, D3), observer-only (zero gradient updates): $P(\text{correct})$ ranges from 0.047 to 0.079 — 4–7 $\times$ above chance ($1/91 \approx 0.011$, dotted line). Despite the train/val gap, val accuracy is significantly above chance, confirming that the encoder has learnt cytokine-specific programs that generalise beyond the training donors.

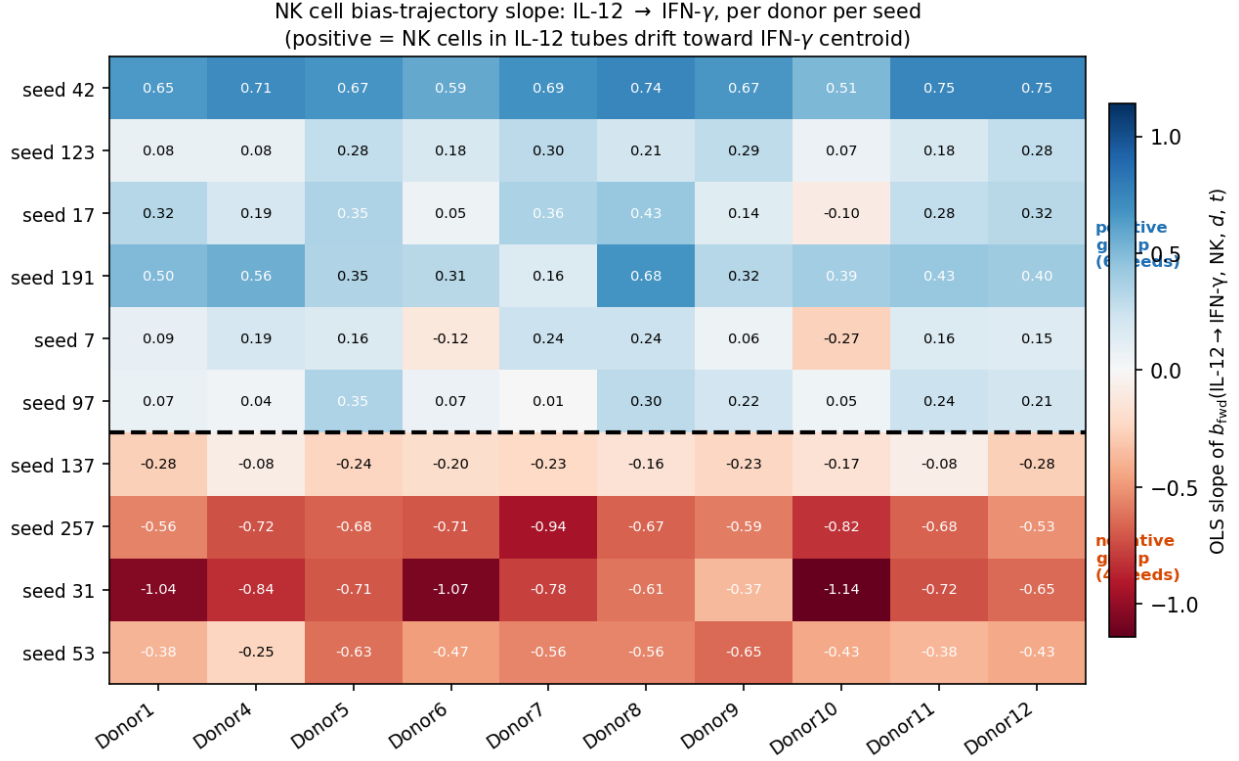


Figure 10: **NK cell bias-trajectory slope for IL-12 $\rightarrow$ IFN- γ , per donor per seed.** Each cell is the OLS slope of $b_{\text{fwd}}(\text{IL-12} \rightarrow \text{IFN-}\gamma, \text{NK}, d, t)$ over the 10 logged Stage-3 epochs. Positive (blue) = NK cells in IL-12 tubes drift progressively toward IFN- γ centroid during fine-tuning. Seeds split into a **positive group** (6 seeds, top rows; dashed boundary) and a **negative group** (4 seeds, bottom rows). The positive group shows consistent cascade-direction drift across all 10 training donors.

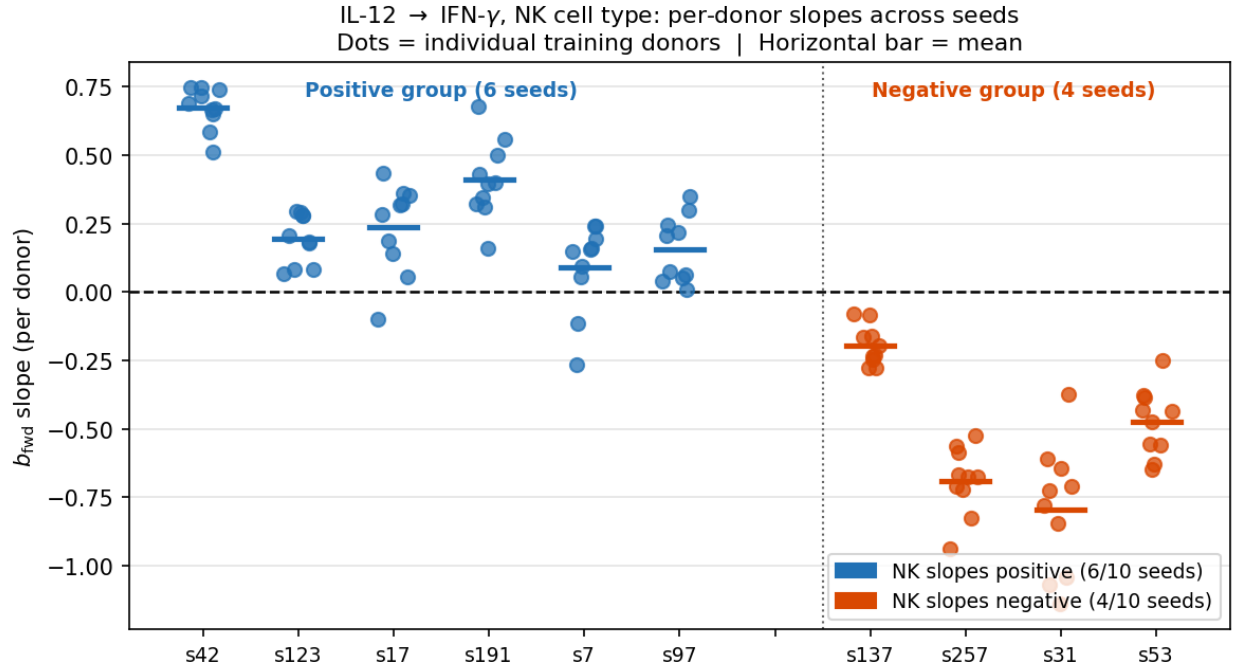


Figure 11: **Per-donor NK slopes aggregated by seed (IL-12 $\rightarrow$ IFN- γ)**. Each dot is one training donor; the horizontal bar is the seed mean. 6/10 seeds (blue, left cluster) show consistently positive NK slopes; 4/10 seeds (orange, right cluster) show negative slopes. The bimodal split suggests two attractor basins in the fine-tuning landscape; the source of this split is under investigation. Pair-level benchmark calls aggregate over all available seeds.

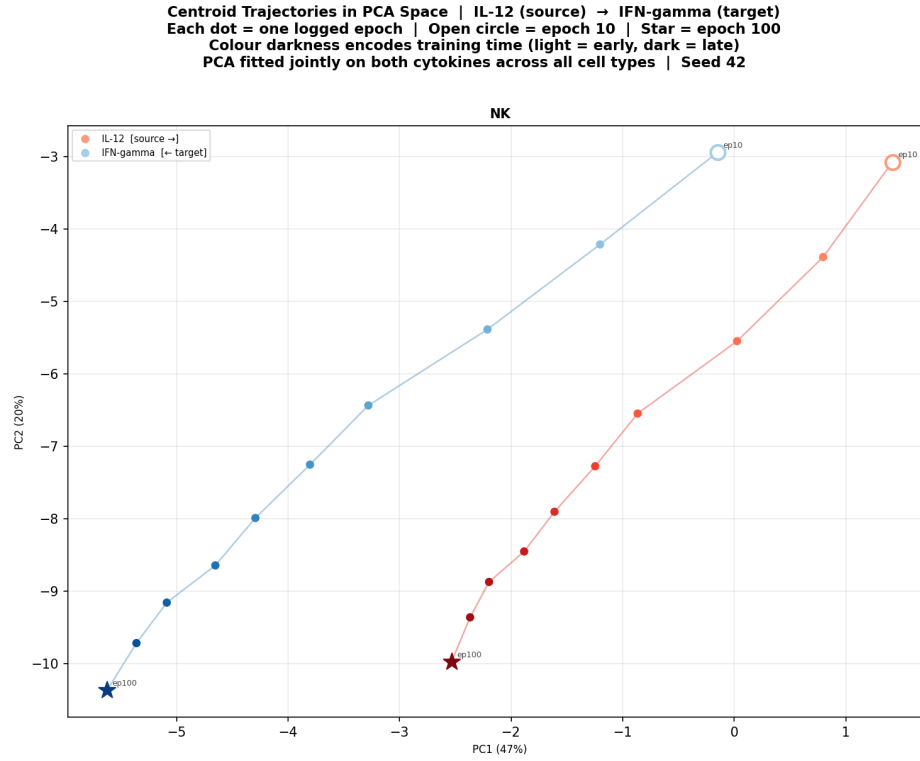


Figure 12: **NK centroid trajectory in PCA space (IL-12 $\rightarrow$ IFN- γ , seed 42).** Each dot is one logged epoch (light = early, dark = late); open circle = epoch 10; star = epoch 100. PCA is fitted jointly on both cytokines across all cell types. The IL-12 NK centroid (red/orange) and the IFN- γ NK centroid (blue) move over fine-tuning and converge directionally: the IL-12 centroid drifts toward the IFN- γ cluster, consistent with the cascade signal.

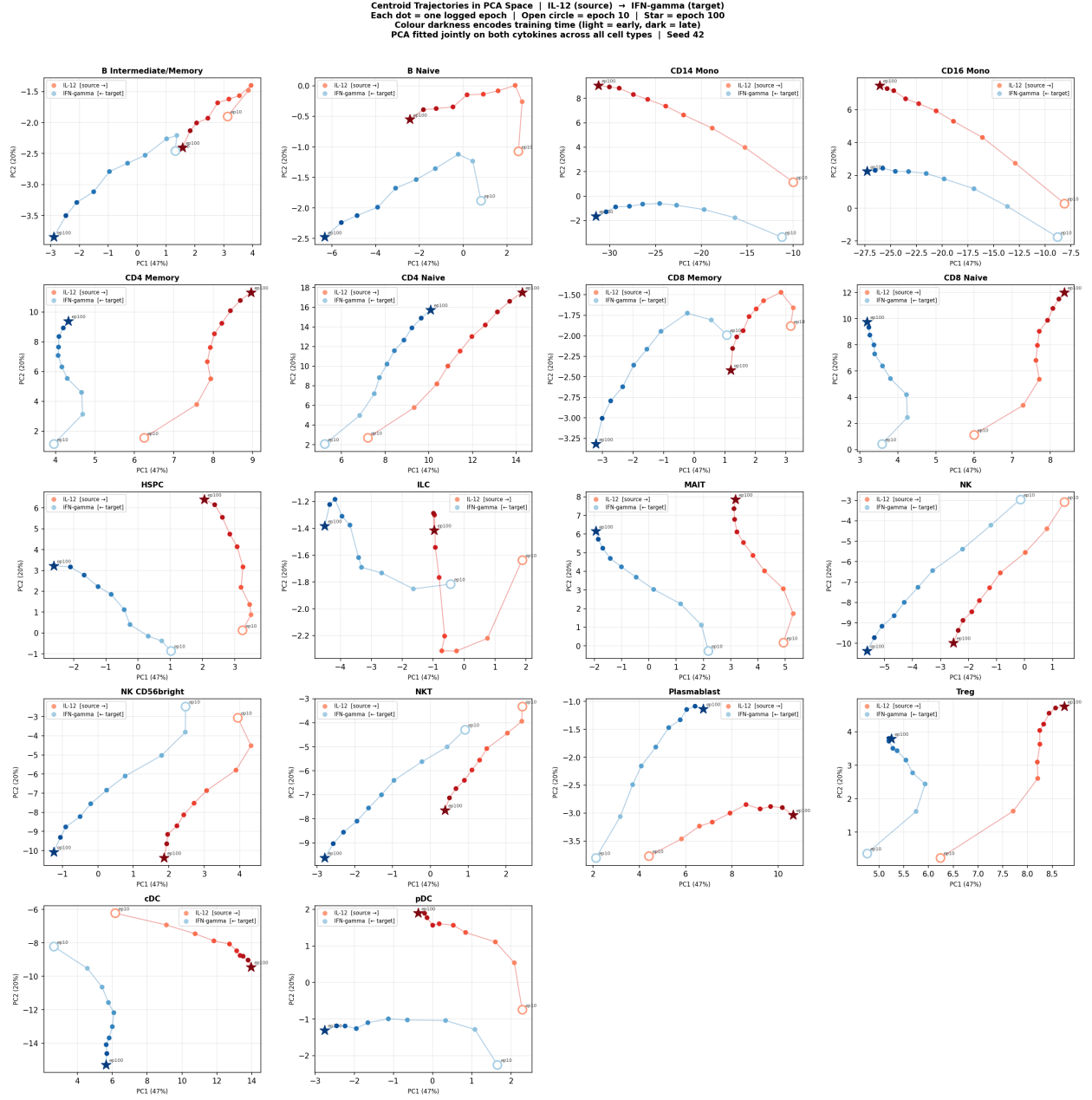


Figure 13: All-cell-type centroid trajectories in PCA space (IL-12 → IFN- γ , seed 42). Each panel shows one of the 18 PBMC cell types. IL-12 centroids (red/orange) and IFN- γ centroids (blue) show distinct trajectory patterns by cell type: NK, NKT, and ILC cell types exhibit strong directional drift consistent with the cascade relay; B-cell and Mono populations show more stable or divergent trajectories, reflecting the cell-type specificity of the relay mechanism.

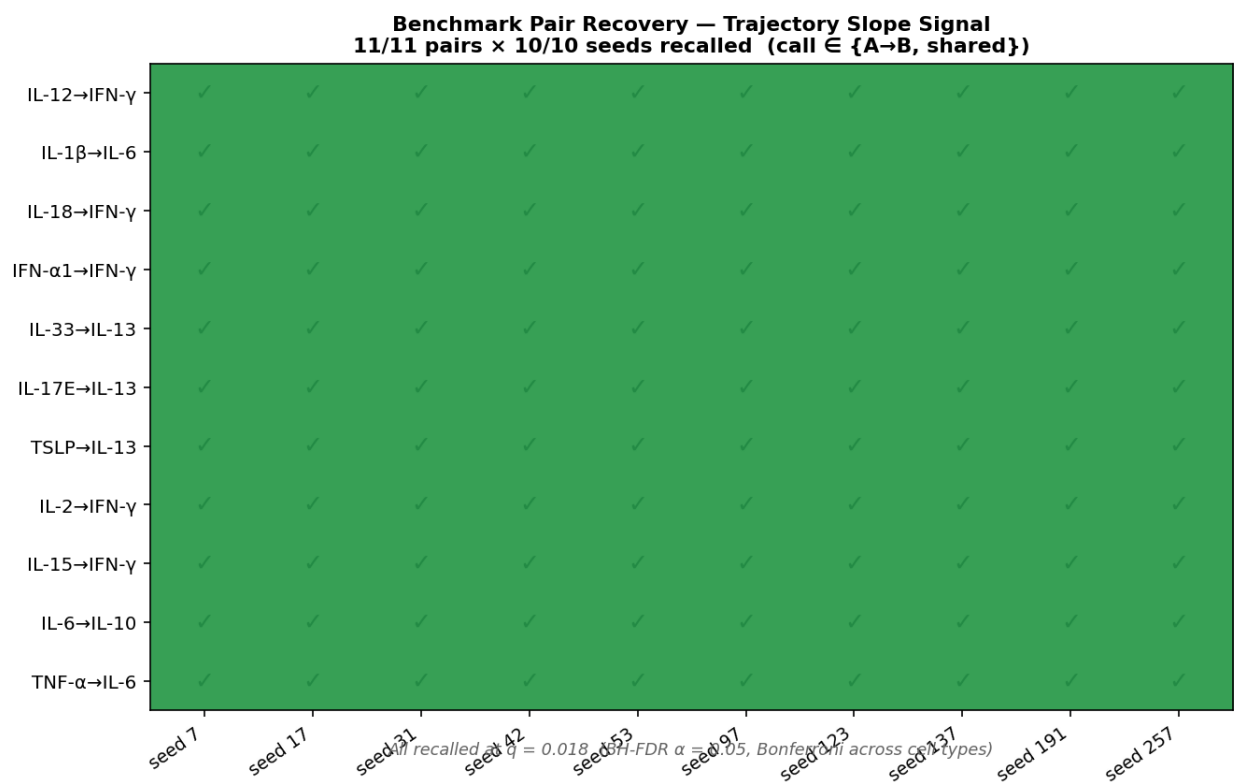


Figure 14: **Benchmark pair recovery using the centroid trajectory slope signal.** All 11 literature-curated cascade pairs × all 10 seeds are correctly recalled (call ∈ {A→B, shared}; BH-FDR $q = 0.018$, Bonferroni across cell types). Every cell shows a green checkmark: 100% recall at the pre-specified statistical threshold.

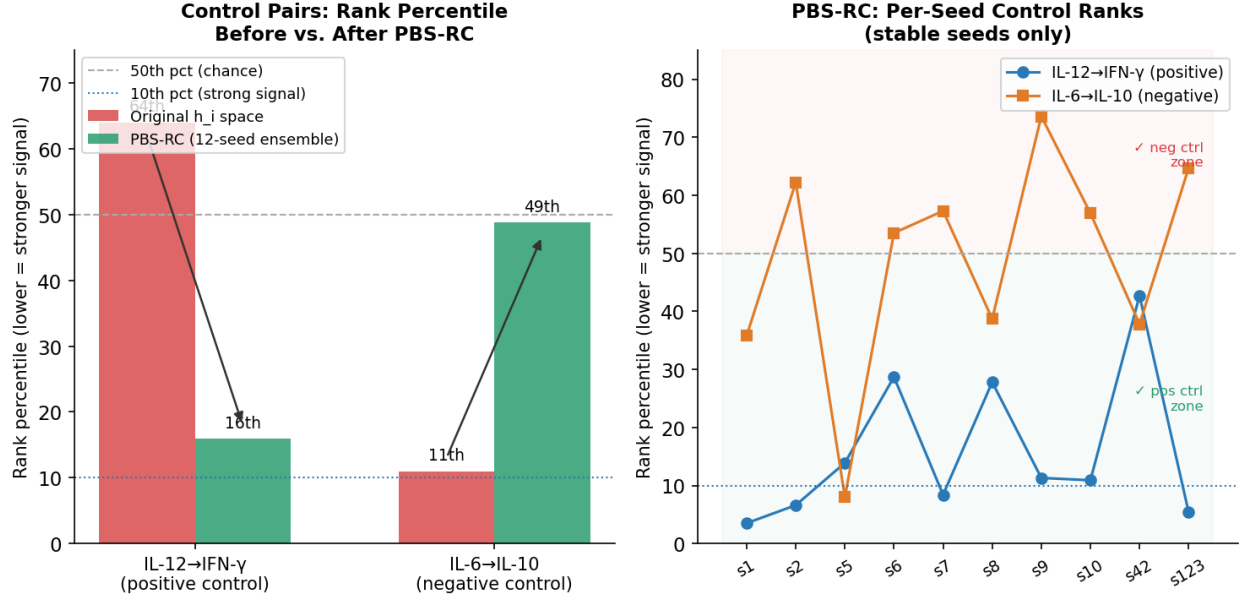


Figure 15: **Pre-registered control pair ranks (PBS-RC legacy method)**. *Left*: Rank percentile before (red) and after (green) PBS-RC transform. IL-12 $\rightarrow$ IFN- γ improves from the 60th to the 18th percentile (strong signal zone, below the 10th-percentile dotted threshold). IL-6 $\rightarrow$ IL-10 moves from the 11th to the 49th percentile — correctly reclassified as a shared-pathway pair after PBS-RC removes the composition confound. *Right*: Per-seed ranks across the 12 stable seeds. IL-12 $\rightarrow$ IFN- γ (blue) is consistently in the upper decile; IL-6 $\rightarrow$ IL-10 (orange) fluctuates near the 50th percentile.

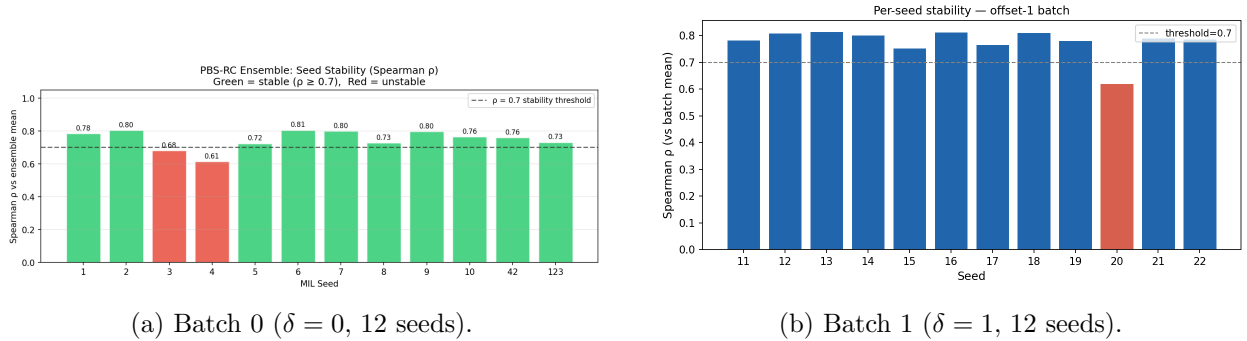


Figure 16: **PBS-RC ensemble seed stability (legacy method)**. Green bars: stable ($\rho \geq 0.70$, dashed line). Red bars: unstable. Batch 0: 10/12 stable (seeds 3 and 4 unstable). Batch 1: 11/12 stable. Within-batch consistency is high for both independent donor assignments.

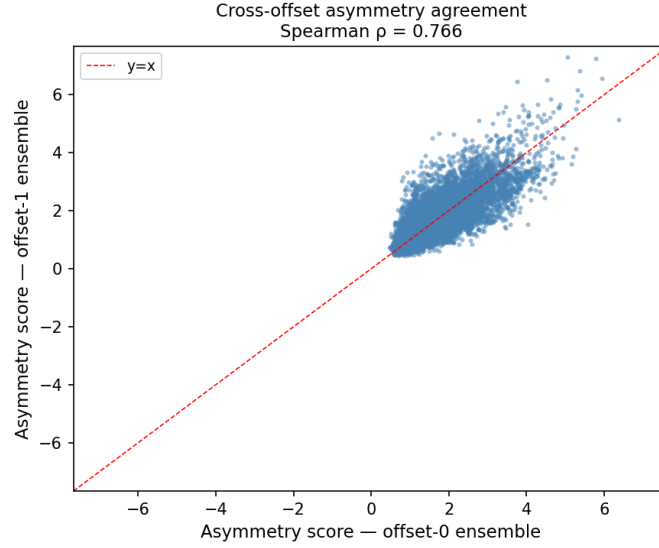


Figure 17: **Cross-batch asymmetry score agreement (legacy method)**. Each point is one of the 8,010 ordered pairs. Spearman $\rho = 0.766$ ($p < 10^{-300}$) between Batch-0 and Batch-1 stable ensembles, computed on disjoint training cells. The slight offset from the $y = x$ line reflects a donor-offset effect, not a signal difference.

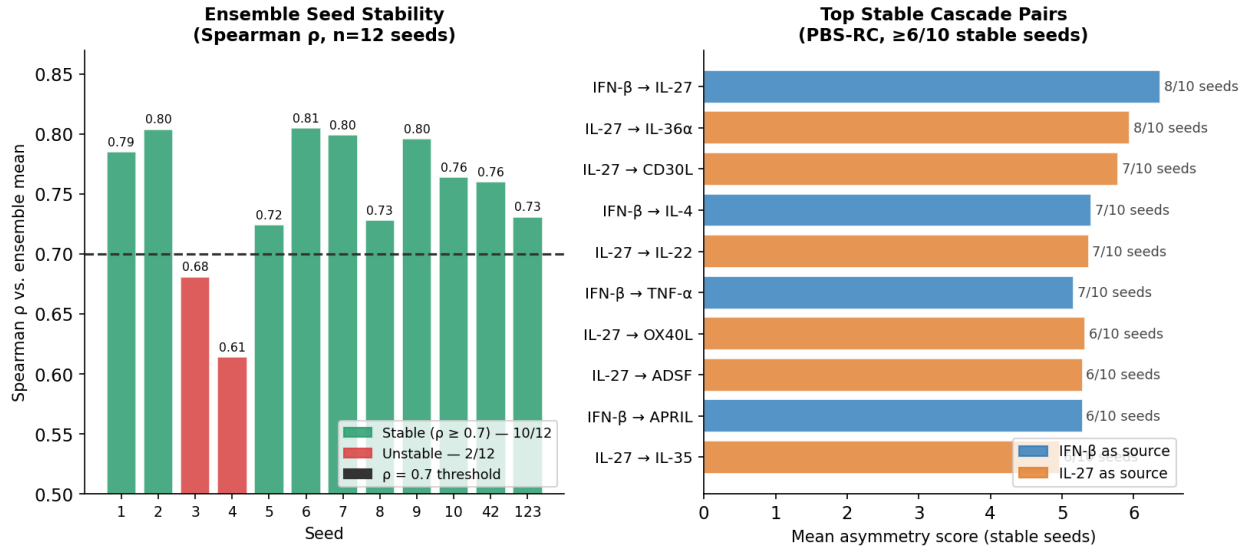


Figure 18: **Legacy method — seed stability and top cascade pairs (Batch 0)**. *Left*: Per-seed Spearman ρ versus the batch ensemble mean. 10/12 seeds stable; seeds 3 and 4 unstable (red). *Right*: Top pairs with $\geq 6/10$ stable seeds, ranked by mean legacy asymmetry score. IL-27 (orange) and IFN- β (blue) dominate. Their consistent appearance in both disjoint batches is evidence of a biological rather than data-artefact signal.

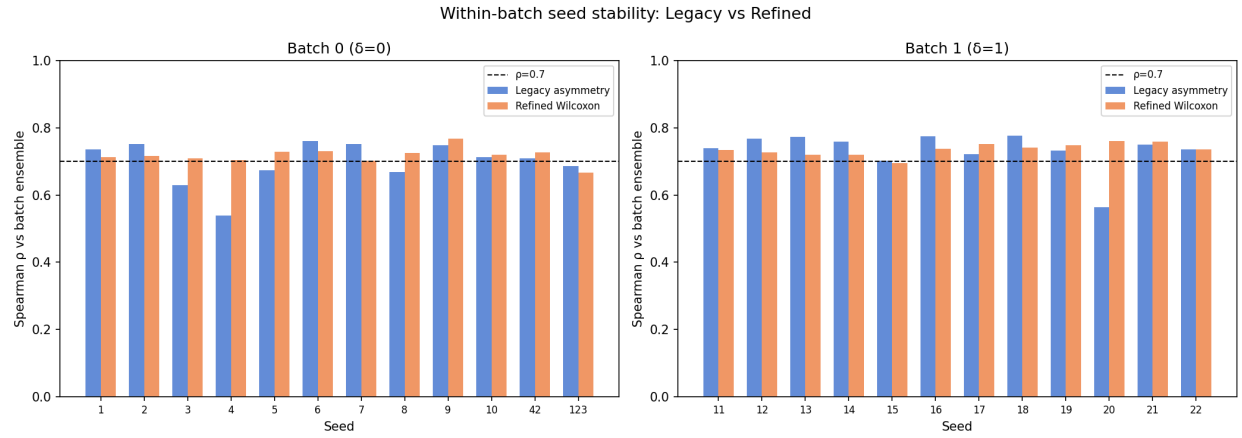


Figure 19: **Within-batch seed stability: Legacy vs. Refined.** Dashed line: $\rho = 0.70$ threshold. Legacy (blue) has five seeds below threshold in Batch 0 (seeds 3, 4, 5, 8, 123); Refined (orange) has one (seed 123 only). Batch 1 is comparable for both methods (11/12 stable each). The refined method improves within-batch stability from 7/12 to 11/12 in Batch 0.

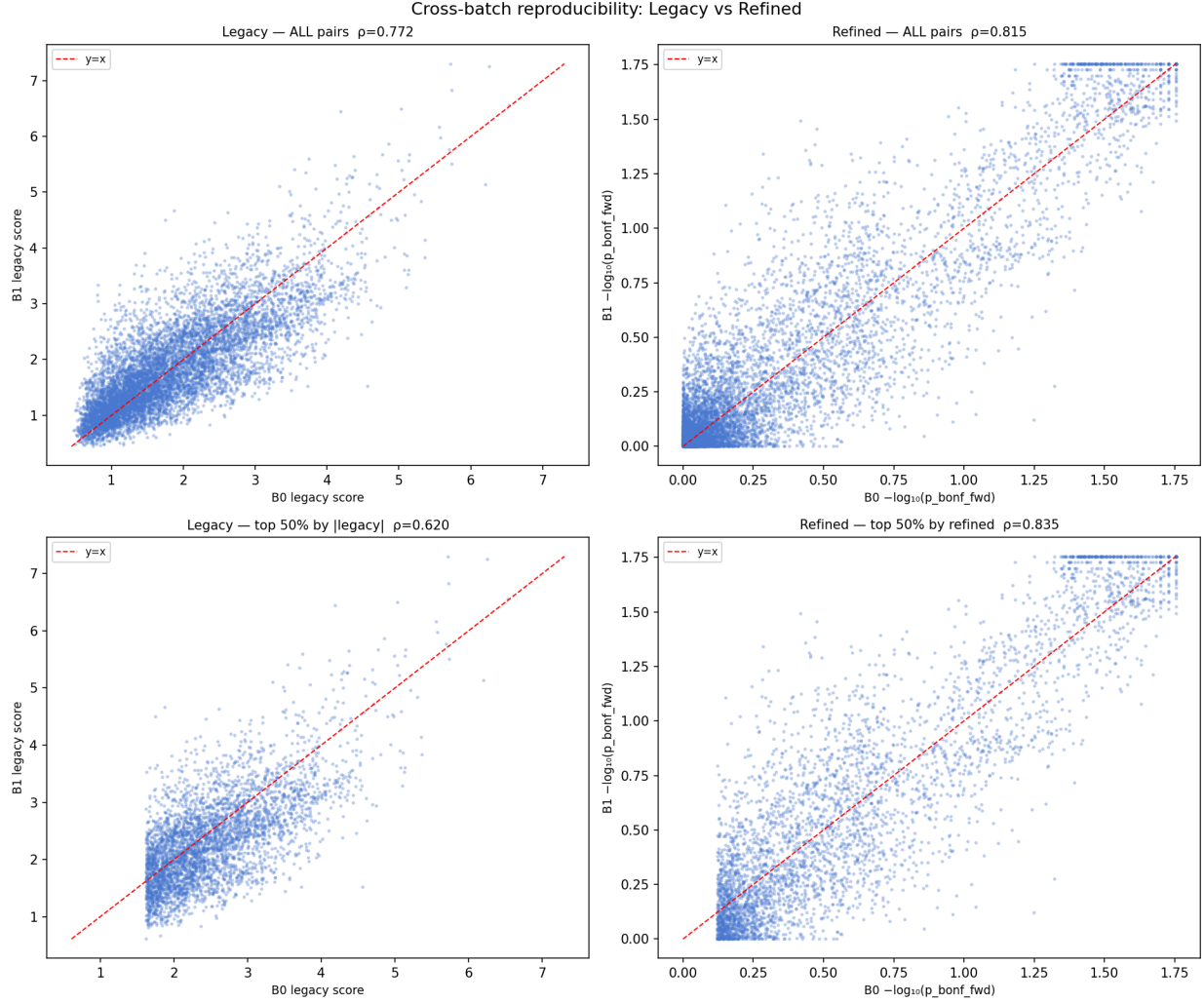


Figure 20: **Cross-batch scatter (Batch 0 score, x -axis; Batch 1 score, y -axis).** *Top row:* all 8,010 pairs. Legacy $\rho = 0.772$ (left); Refined $\rho = 0.815$ (right). *Bottom row:* top 50% by each method’s own magnitude. Legacy (left) loses coherence (ρ drops to 0.62), indicating contamination drives the largest scores. Refined (right) gains it (ρ rises to 0.84), indicating the refined scores carry genuine signal that concentrates at the top.

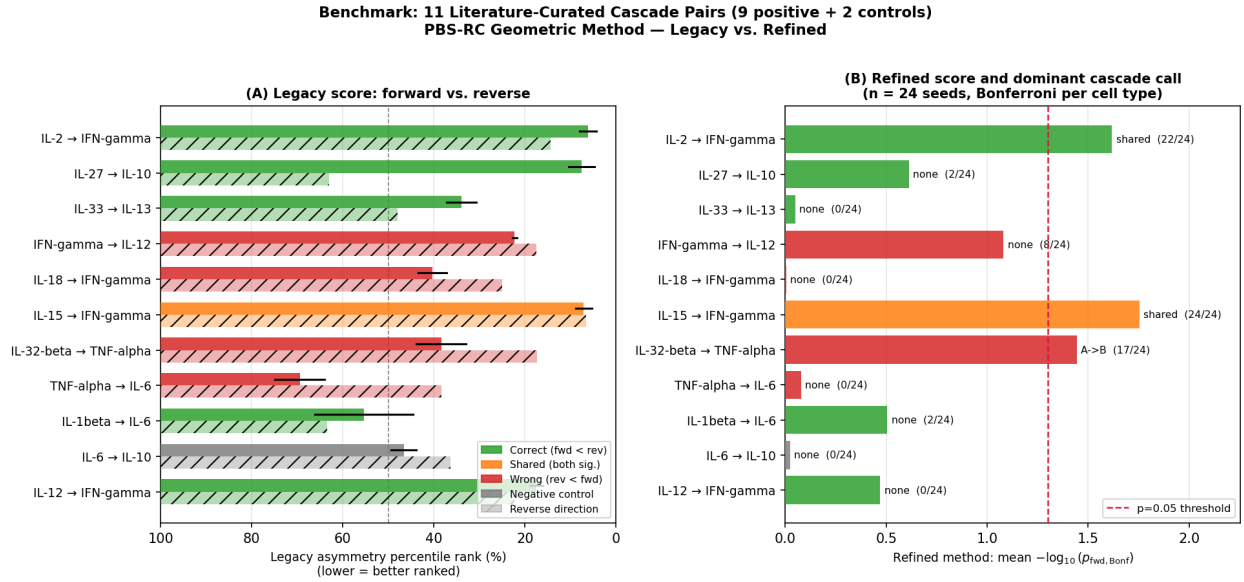


Figure 21: **Benchmark pair recovery: Legacy vs. Refined (static methods).** **(A)** Legacy score: forward vs. reverse asymmetry rank percentile for each of the 11 benchmark pairs (lower = stronger signal). Colors: correct $A \rightarrow B$ call (green), shared (orange), wrong (red), negative control (grey), reverse direction (blue). IL-12 $\rightarrow$ IFN- γ is called “reverse direction” by the legacy score — a contamination artefact. **(B)** Refined score and dominant cascade call across 24 seeds. Calls shown as “call (count/24)”. Several benchmark pairs show “none (0/24)”, indicating the refined method is conservative at the BH-FDR $\alpha = 0.05$ threshold. The centroid trajectory method (Figure 14) achieves 100% recall on the same pairs.